Exact Realizer Floors for One- and Two-Defect Valuation Words in the Accelerated $3x + 1$ Map

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Abstract

For the accelerated Syracuse map $T(x) = (3x + 1)/2^{v_2(3x+1)}$ on odd integers we give the exact least realizer of a two-defect valuation word $w = 1^u q_a 1^v q_b 1^t$ in closed form, and an unconditional per-configuration floor $F_c(w)$ obtained by combining truncation with orbit-advance at one-defect granularity. We record the modulus convention precisely: the exact-valuation realizer set is one residue class modulo 2^{S+1}, not 2^S. We prove that the anchor denominator is 3^n on the interior stratum $u, v, t \geq 1$ and identify the boundary stratum $t = 0$ where it drops, the extreme witness being the $x = 1$ cycle word. We then sharpen the floor to F_c' using the fact that the second defect closes the first episode, and exhibit an infinite family on which that sharpening gains an unbounded number of bits. We then reduce the *uniform* two-defect floor to a single named one-defect import (β), with the threshold $\beta > 1/2$ exact at every scale, and record two negative results with computed witnesses.

Non-claims, stated up front. Nothing here bears on the Collatz conjecture and nothing excludes divergent orbits; every statement concerns least realizers of finite valuation words. There is no formal verification: "verified" means exhaustive or exact finite computation on a stated grid. Theorem 5 is a per-configuration certificate and is *not* a uniform family floor. The hypothesis (β) is open. A limited prior-art audit was carried out (§9) but is not exhaustive, so novelty is not established; branches (a) and (b) of Theorem 5 are standard cylinder facts and Proposition 1 restates a correction published earlier by the author. What is claimed here is the closed-form per-configuration evaluation, the closed-episode branch (c) with its unbounded gain (Proposition 6), and the β-reduction.

1 Epistemic tiers

Tier	Meaning	Used for
T1	Exact integer arithmetic; proof	Props. 1–2, Thms. 3, 5, Lem. 4, all grid counts
T2	Exact integers, `math.log2` on exact inputs	δ values (§7); Pearson estimates (§10)
T3	Interpretation	framing remarks, §10 items 4–6

`mpmath` was unavailable in the working environment, so the T2 tier is served by exact integers rather than arbitrary-precision floats, which is stronger than required. No floating-point value decides any assertion.

2 The realizer class and the modulus convention

Throughout $\alpha = \log_2 3 = 1.5849625\ldots$. A valuation word is $w = (v_1, \ldots, v_n)$ with $v_k \geq 1$; set $S_k = v_1 + \cdots + v_k$, $S = S_n$, $|w| = n$. The affine identity is $2^{S_k} x_k = 3^k x + C_k$ with append recurrence $C' = 3C + 2^S$, equivalently $C_w = \sum_{k=1}^{n} 3^{n-k} 2^{S_{k-1}}$.

Proposition 1. *An odd m realizes w with exact valuations if and only if $3^n m + C_w \equiv 2^S \pmod{2^{S+1}}$. The realizer set is exactly one residue class modulo 2^{S+1}; write $m(w)$ for its least positive member.*

Proof. Two steps. (a) The top divisibility $2^{S_n} \mid 3^n m + C_n$ implies every earlier one: from $C_n = 3C_{n-1} + 2^{S_{n-1}}$, $3^n m + C_n = 3(3^{n-1}m + C_{n-1}) + 2^{S_{n-1}}$, and $2^{S_{n-1}}$ divides the left side and the trailing term, hence $3(3^{n-1}m + C_{n-1})$ and so $3^{n-1}m + C_{n-1}$. The same computation gives $2^{v_n} \mid 3x_{n-1} + 1$, so each x_k with $k < n$ is automatically odd. (b) The one remaining condition is that x_n be odd, worth exactly one further bit; it upgrades the integrality modulus 2^S to 2^{S+1}. □

The *integrality* class alone is modulo 2^S; the exact-valuation class is modulo 2^{S+1}. The distinction is the difference between a realizer and the least realizer, and it is the source of the reconciliation in §3.

Verification. An independent implementation sharing no code with the first builds the class by backward recursion, $x \equiv 3^{-1}(2^{v_1}a' - 1) \pmod{2^{v_1+M'}}$, from the base "the empty word is realized by every odd x". The modulus 2^{S+1} emerges with no input, and a full scan of *every* odd $m < 2^{S+1}$ on 243 words confirms a singleton class with $M = S + 1$ in every case: 0 failures.

3 One defect, and a notation reconciliation

The conjugation $y = x + 1$ linearises a run of ones: $v_k = 1$ gives $x_k + 1 = 3(x_{k-1} + 1)/2$, so $x_k + 1 = (3/2)^k(x_0 + 1)$. A leading run of u ones forces $2^u \mid m + 1$, and a following letter $q \geq 2$ forces $2^{u+1} \parallel m + 1$.

Proposition 2. *Let $m + 1 = 2^{u+1}A$ with A odd, $n = q + v$, and let $y = (3^{u+1}A - 1)/2^{q-1}$ be the odd orbit value reached after $1^u q$. Then the run of v ones is governed by $v_2(y + 1)$, and the two natural conditions are* distinct*:*

condition on the orbit	*meaning*	*congruence*
$v_2(y+1) \geq v+1$	*realizes the finite word $1^u q 1^v$*	$3^{u+1}A \equiv 1 - 2^{q-1} \pmod{2^n}$
$v_2(y+1) = v+1$	*… and the next valuation is ≥ 2*	$3^{u+1}A \equiv 1 - 2^{q-1} + 2^n \pmod{2^{n+1}}$

Write $m(P)$ for the least realizer under the first condition and $m_{\text{closed}}(P)$ for the least under the second.

Proof. After 1^u the orbit is at $2 \cdot 3^u A - 1$, so $3x + 1 = 2(3^{u+1}A - 1)$ and $v_2 = q$ forces $v_2(3^{u+1}A - 1) = q - 1$, giving $x_{u+1} = y$. A run of v ones needs $2^v \mid y + 1$ (conjugate $y \mapsto y + 1$, under which the run is $y \mapsto 3y/2$), and the final value $3^v(y + 1)/2^v - 1$ is odd exactly when $2^{v+1} \mid y + 1$. For the next valuation, $3x_n + 1 = 2\big(3^{v+1}(y + 1)/2^{v+1} - 1\big)$, which is divisible by 4 precisely when $2^{v+1} \parallel y + 1$. □

The source's finer modulus is not an error. The imported note states the mod-2^{n+1} form. That is the *closed episode* condition, a different and equally legitimate specification—the end of a whole episode rather than an arbitrary finite prefix—not a typo for 2^n. Both are used below: Proposition 2's first row for the general theory, its second row in Theorem 5 (c).

The two realizers can differ by many bits. $m_{\text{closed}}(P) \in \{m(P),\, m(P) + 2^{S_P+1}\}$, so the two differ by at most one *modulus*—but their *heights* can differ by far more than one bit, since $m(P)$ may sit well below 2^{S_P+1}. At $u = 11$, $q = 5$, $v = 4$: $m(P) = 4095$ while $m_{\text{closed}}(P) = 2\,101\,247 = 4095 + 2^{21}$, so $\log_2\big(m_{\text{closed}}/m\big) = 9.0032$. The measured supremum grows with the grid (7.4237 for $u, v \leq 10$; 9.0032 for $u, v \leq 14$; 9.4294 for $u, v \leq 18$), so this

is not an $O(1)$ correction. One extra congruence bit is not one extra bit of height, and any estimate comparing the two must use the height, not the modulus count.

Placement reduction. Minimality of A is governed by where $c\,3^{-(u+1)}$ falls in $\mathbb{Z}/2^n$; dually, by the ternary digits of 2^{-n} modulo 3^{u+1}. This is the one-dimensional case of the obstruction in §10.

4 Two defects: closed form and denominator

Let $w = 1^u q_a 1^v q_b 1^t$, $n = u+v+t+2$, $S = u+v+t+q_a+q_b$.

Theorem 3. $C_w = 3^n - 3^{\,v+t+1}2^{\,u+1} + 3^{\,v+t+1}2^{\,u+q_a} - 3^{\,t}2^{\,u+q_a+v+1} + 3^{\,t}2^{\,u+q_a+v+q_b} - 2^S$.

Equivalently, by Proposition 1,

$$m(w) \equiv -1 - 3^{-(u+1)}2^{u+1}\big(2^{q_a-1}-1\big) - 3^{-(u+v+2)}2^{\,u+q_a+v+1}\big(2^{q_b-1}-1\big) \pmod{2^{S+1}},$$

which exhibits $2^{u+1} \parallel m+1$, both cofactors being odd. At $q_a = q_b = 1$ the formula collapses to $3^n - 2^n$, the all-ones value. Verified: 0 mismatches out of 19 773 against the append recurrence.

Lemma 4. *The anchor is $3^{-n}(2^S - C_w)$, with reduced denominator $D = 3^{\max(0,\, n - v_3(2^S - C_w))}$. On the interior stratum $u, v, t \geq 1$, $v_3(2^S - C_w) = 0$ and hence $D = 3^n$.*

Proof. With $t \geq 1$ every monomial of C_w except -2^S carries a factor 3, so $C_w \equiv -2^S \pmod 3$ and $2^S - C_w \equiv 2^{S+1} \not\equiv 0 \pmod 3$. □

The denominator is representative-dependent. Both one-sided lifts pin one representative each; the canonical Lemma-H datum is the *balanced* lift $N^* \equiv -C \pmod{2^S}$ with $|N^*| \leq 2^{S-1}$, giving $D = 3^{\,n - v_3(N^*)}$. On the family $u, v, t \in 1..12$ with $q_a, q_b \in \{2,3,4\}$ (15 552 configurations), verified independently: both one-sided representatives are 3-coprime everywhere (0/15 552 each); the balanced lift reduces on 5263/15 552 = 0.3384, with $v_3(N^*)$ following a geometric(2/3) law to within sampling error:

k	0	1	2	3	4	5	6	7	8	9
observed	10289	3478	1199	417	123	32	9	3	1	1
geom(2/3)	10368	3456	1152	384	128	42.7	14.2	4.7	1.6	0.5

The dangerous-window probe gives median $\log_2 m/(S+1)$ of 0.9615 at $v_3(N^*) = 0$ versus 0.9620 at $v_3(N^*) > 0$, a difference of 0.00046. On these grids the reduction is therefore **consistent with the geometric benchmark and shows no predictive signal about** m. This is a finite measurement, not a proof that no relationship exists.

The premise $\gcd(C_w, 3^n) = 1$ holds on all 19 773 configurations but does *not* imply $D = 3^n$, since D is governed by $\gcd(2^S - C_w, 3^n)$. Stratifying over $|w| \leq 44$, reductions $v_3 > 0$ occur on 1176 configurations with $t = 0$ and on 0 with $t \geq 1$: the reduction is confined to $t = 0$ exactly. The truncation at 0 is necessary, not cosmetic: v_3 may exceed n. For $w = (6,2)$, $n = 2$, $S = 8$, $C_w = 67$ and $2^S - C_w = 189 = 3^3 \cdot 7$, so $v_3 = 3 > n$; the anchor is $189/9 = 21$ with reduced denominator 1, not 3^{-1}. (On a grid with $u, v, t \leq 6$ and $q_a, q_b \in \{2, \ldots, 7\}$ this occurs on 4/12 348 configurations, all with $v = t = 0$; it is invisible on a grid restricted to $q_a, q_b \in \{2,3,4\}$.) The extreme witness is the $x = 1$ cycle word $w = (2,2)$, where $m(w) = 1$, $2^S - C_w = 9$, $v_3 = 2$ and $D = 1$: reduced denominators mark cycle proximity, consistent with the cycle-denominator identification of Lagarias [11]. Accordingly the family is stated as $u, v, t \geq 1$ and the $t = 0$ and $v = 0$ strata are carved out explicitly.

5 The suffix-combination floor

Let $P = 1^u q_a 1^v$, $P' = 1^u q_a$, $Q = 1^v q_b 1^t$.

Theorem 5. *(a)* $m(w) \geq m(P)$; *(b)* $m(w) \geq \lceil (2^{S_{P'}} m(Q) - C_{P'})/3^{u+1} \rceil$; *(c)* $m(w) \geq m_{\text{closed}}(P) \geq m(P)$. *Hence* $F'_c(w) := \max\{(b),(c)\}$ *is an unconditional floor, computable in closed form for each configuration, and* $F'_c \geq F_c := \max\{(a),(b)\}$.

Proof. (a) Truncation: any realizer of w realizes the prefix P, so $m(w)$ lies in the class of P. *(b) Orbit-advance:* x_{u+1} realizes Q, so $x_{u+1} \geq m(Q)$, and $m = (2^{S_{P'}} x_{u+1} - C_{P'})/3^{u+1}$ is increasing in x_{u+1}. *(c)* In $w = 1^u q_a 1^v q_b 1^t$ the prefix P is followed by $q_b \geq 2$, so every realizer of w realizes P *as a closed episode*; apply the second row of Proposition 2. □

The closed branch is a strict improvement. Branch (c) is available only because the second defect closes the first episode—it is not an alternative convention but extra information the two-defect structure supplies for free. Measured on $u, v, t \in 1..11$ (15 552 configurations): F'_c is *never* worse than F_c and is strictly better on 5442 (35.0%), with mean gain 1.649 bits, median 1.414, maximum 4.790. Truncation binds in 64.2% of configurations. On that grid the improvement is small—the balanced median rises from 0.6826 to 0.6903 per unit $|w|$ against an empirical 1.0341—but **it is not** $O(1)$ **in general**, and an earlier draft's claim that it was has been withdrawn.

Proposition 6. *The gain* $\log_2(F'_c/F_c)$ *is unbounded on the two-defect family. For each* $v \geq 1$ *choose an even* e *with* $3^e \equiv -15 \pmod{2^{v+6}}$ *and* e *large enough that branch (b) is dominated. For* $w = 1^{e-1} 5\, 1^v 2\, 1$,

$$F_c(w) = 2^e - 1, \qquad F'_c(w) = 2^e - 1 + 2^{e+v+5}, \qquad \log_2 \frac{F'_c(w)}{F_c(w)} > v + 5.$$

Proof. Existence of arbitrarily large even e. For $k \geq 3$ the group $(\mathbb{Z}/2^k)^*$ is $\{\pm 1\} \times \langle 3 \rangle$ with $\operatorname{ord}(3) = 2^{k-2}$, so $\langle 9 \rangle = \langle 3^2 \rangle$ has order 2^{k-3} and is contained in $\{x \equiv 1 \bmod 8\}$, a set of the same cardinality; hence $\langle 9 \rangle = \{x \equiv 1 \bmod 8\}$ exactly. Since $-15 \equiv 1 \pmod 8$, the congruence $3^e \equiv -15 \pmod{2^{v+6}}$ is solvable, and because $3^e \equiv 1 \pmod 8$ iff e is even, every solution is even. The solutions form the progression $e \equiv e_0 \pmod{2^{v+4}}$, so arbitrarily large even solutions exist.

The two floors. Put $u = e - 1$, $q_a = 5$, so $P = 1^{e-1} 5\, 1^v$, $n = q_a + v = v + 5$ and $S_P = e + v + 4$. Taking $A = 1$ in Proposition 2 gives $m = 2^e - 1$, and the first row requires $3^e \equiv 1 - 2^4 = -15 \pmod{2^{v+5}}$, which our congruence implies; as $A = 1$ is the least positive odd value, $m(P) = 2^e - 1$. The second row requires $3^e \equiv -15 + 2^{v+5} \pmod{2^{v+6}}$, which our congruence *contradicts*; so $A = 1$ is not closed, and $m_{\text{closed}}(P) = m(P) + 2^{S_P+1} = 2^e - 1 + 2^{e+v+5}$. This is exactly why the hypothesis is imposed modulo 2^{v+6} and not 2^{v+5}: the finer modulus simultaneously selects $A = 1$ for the open word and rejects it for the closed one.

Branch (b) is dominated. With $t = 1$, $Q = 1^v 2\, 1$ and $m(Q)$ depend only on v, while $S_{P'} = u + q_a = e + 4$. Hence branch (b) is at most $2^{e+4} m(Q)/3^e$, and its ratio to $2^e - 1$ is $O\big(16\, m(Q)\, 3^{-e}\big) \to 0$. So for all large e in the progression the truncation branch binds in both F_c and F'_c, giving $\log_2(F'_c/F_c) = \log_2\big(1 + 2^{v+5} \cdot \frac{2^e}{2^e - 1}\big) > v + 5$. □

Each step was checked numerically: $\langle 9 \rangle = \{x \equiv 1 \bmod 8\}$ for $2^4 \ldots 2^{12}$; the solution progressions and their parity for $v = 1, \ldots, 5$; the open/closed selection of $A = 1$; and the domination of branch (b) along the progression. Verified for $v = 1, \ldots, 7$: both closed forms match the computed floors exactly, and the gain is $6.0227, 7.0116, 8.0060, 9.0032, 10.0014, 11.0007, 12.0004$, decreasing to $v + 5$ from above. The smallest instance is already instructive: $w = 1^{11} 5\, 1^4 2\, 1$ has $F_c = 4095$, $F'_c = 2\,101\,247$ and $m(w) = 14\,684\,159$, a gain of 9.0032 bits with both floors valid. Branch (c) therefore sharpens the per-configuration certificate by an unbounded amount

on suitable words, while leaving §6 and every asymptotic claim untouched (those concern the balanced regime, where the gain is small).

Framing (T3). F_c is **not a new mechanism**: it is orbit-advance plus truncation, evaluated at one-defect granularity—truncating to the full one-defect prefix P and using its exact $m(P)$ rather than to a maximal all-ones run. The novelty claimed is only closed-form per-configuration evaluability and the β-reduction of §6.

Independence certificate. $F_c \le m(w)$ with 0 violations out of 19 773, and $F_c' \le m(w)$ with 0 out of 15 552. Against the all-ones-run floor: better on 19 252, tied on 521, worse on 0; dominant, strictly better on 97.4%. The second, independent implementation of the same specification reports 450/451 with one exact tie at a degenerate $|w| = 32$, $v = 1$ configuration—the same qualitative result on a different grid. That package was run in this environment and reproduced its balanced medians 0.337/0.671/1.015 exactly, against 0.683/0.336/1.034 here; the differences are grid composition.

6 The uniform reduction

Theorem 5 is per-configuration. A uniform family floor needs a uniform one-defect input.

Hypothesis β. $\log_2 m(1^U q 1^V) \;\ge\; \beta \cdot S(1^U q 1^V)$ for all $U \ge 1$, $V \ge 1$, $q \ge 2$.

The domain restriction is necessary. With $V = 0$ allowed the hypothesis is false for any $\beta > 1/2$: the word $(1, 4)$, i.e. $U = 1$, $q = 4$, $V = 0$, has $m = 3$ and $S = 5$, giving $\log_2 m/S = 0.3170$. Allowing $U = V = 0$ is worse still—$w = (2)$ has $m = 1$ and ratio 0. On the interior $U, V \ge 1$, with $q \in \{2, \dots, 7\}$, the measured minimum is 0.5714, attained at $(U, q, V) = (11, 5, 5)$, and it is *stable*: the same minimiser and value on grids $U, V \le 20$, ≤ 30 and ≤ 45 (0.5977 for $U, V \le 10$). So the interior family is consistent with $\beta > 1/2$, with a margin of about 0.07 and no observed drift toward 1/2—evidence for the hypothesis, not a proof of it. The two-defect application only ever invokes Hypothesis β on the interior, since $P = 1^u q_a 1^v$ with $u, v \ge 1$.

Normalisation, stated explicitly: balanced $u = v = t = k$, $q_a = q_b = 2$, so $S_P = 2k + 2$, and the comparator is the all-ones-run floor of $k + 1$ bits.

- *Truncation branch:* $\beta(2k + 2) > k + 1 \iff \beta > (k + 1)/(2k + 2) = 1/2$, exactly 1/2 for *every* k, not merely asymptotically.
- *Advance branch:* $\beta > (\alpha(k + 1) - 1)/(2k + 2) \to \alpha/2 = 0.792481$, rising $0.542 \to 0.792$ across $k = 1, \dots, 5000$.

Empirically β for $1^k 21^k$ is 0.9167 ($k = 5$), 0.9286 (20), 0.9672 (60), 1.0033 (150). **Hypothesis β with $\beta > 1/2$ is the single question to which this note reduces the uniform two-defect problem. It is open, and it is an effective question.**

7 Negative results

Merge bridge: failed. Canonical image $m_2(u, q_a+q_b, v+t)$. At $(u, q_a, v, q_b, t) = (22, 4, 2, 2, 6)$, $|w| = 32$:

image	bits	δ (bit length)	δ (true $\log_2$, T2)
canonical $m_2(u, q_a + q_b, v + t)$	37	+0.3125	**+0.3209**
$m_2(u, q_a, v + t)$	34	+0.2188	+0.2248
$m_2(u, q_a + q_b - 1, v + t)$	35	+0.2500	+0.2508

The canonical variant reproduces the second implementation's $\delta = 0.321$ to four decimals, provided δ is measured by true $\log_2$ rather than bit length; that cross-engine agreement is the engines-agree certificate. Suprema of δ per unit $|w|$: $+0.8662$ at $(2, 2, 1, 4, 1)$, $|w| = 6$ on the small full grid and on its interior; $+0.2093$ at $(21, 2, 21, 3, 0)$, $|w| = 44$ for $|w| \geq 32$. The small-grid suprema are attained at the smallest admissible words and are boundary artefacts, so the $|w| \geq 32$ figure is the one bearing on asymptotics.

Deletion bridge: dominated by Theorem 5—it requires $\beta \gtrsim 0.56$, strictly worse than $1/2$, and additionally needs a uniform-cost hypothesis that F_c does not.

Why no bridge beats F_c (T3). A bridge posits $m(v) \leq A\, m(w)^q 2^{\delta|w|}$, which rearranges to $m(w) \geq \big(m(v)/(A2^{\delta|w|})\big)^{1/q}$—*the bridge's own conclusion.* So that condition is never weaker than the floor it produces, and a bridge is non-circular only when it is established by a mechanism indifferent to the size of $m(w)$. Exactly one such mechanism exists: a realizer realizes every contiguous factor of w, directly or after orbit advance. Those are the two branches of F_c. Merge and deletion images are not factors of w.

8 Calibration

Medians of bits per $|w|$, balanced regime, $|w| \geq 62$, 369 configurations: all-ones-run floor 0.336; F_c 0.683; F_c' **0.690**; empirical 1.034. Per regime at $|w| \approx 122$ (bits):

regime	trunc	adv	F_c	run floor	empirical	binds
balanced	82	59	**82**	41	124	truncation
leading-heavy	122	1	**122**	119	122	truncation
middle-heavy	119	122	**122**	119	125	advance
trailing-heavy	5	119	**119**	119	124	advance

F_c is near-sharp in the lopsided regimes (0.93–1.00 of the empirical exponent) and loses about a third of it when balanced. Under Hypothesis β with $\beta \approx 0.95$ the conditional balanced rate is ≈ 0.633 per unit word length, so (β) does *not* by itself close the balanced gap; it closes the gap to the run-floor comparator.

9 Relation to prior work

A literature check was performed for this revision. It is not exhaustive, and absence from it is not evidence of novelty.

The affine identity is classical. $2^{S_k} x_k = 3^k x + C_k$ with $C_w = \sum_k 3^{\,n-k} 2^{S_{k-1}}$ is Proposition 4 of Böhm and Sontacchi [3] (1978), in the two-branch form $u^n(x) = \big(3^m x + \sum_{k<m} 3^{\,m-k-1} 2^{v_k}\big)/2^n$; their Proposition 5 is the cycle closed form. *Verified first-hand.*

Proposition 1 is the author's own prior correction. That the exact realizers of a valuation word form a single residue class modulo 2^{S+1} rather than the Terras–Everett modulus 2^S was established in [5] (11 August 2026), explicitly as an erratum to the Terras–Everett residue class, and is carried as Proposition 5.2 of [4]. Proposition 1 restates it here, with a proof, only because the 2^S versus 2^{S+1} distinction drives §3; no new priority is claimed for it in this note.

The same statement appears independently in Akhtar [1] (19 August 2026) as Theorem 9, with the explicit class

$$r(k) \;\equiv\; -\sum_{j=0}^{m} 2^{K_j} 3^{-(j+1)} \pmod{2^{K+1}}, \qquad K_j = \textstyle\sum_{i<j} k_i,$$

derived from the parity-vector bijection. *Full text consulted.* That expression and the form used here, $3^{-n}(2^S - C_w) \bmod 2^{S+1}$, are equivalent: they differ by $2^{K+2} \cdot (\text{odd}) \equiv 0$, verified on 1944 two-defect words with 0 disagreements.

Akhtar's concerns are otherwise disjoint from §5–§6. He uses the least nonnegative representative only in normalized form, r/M with $M = 2^{K+1}$, as a placement diagnostic (his Table 4); he does not bound least realizers from below, give closed forms for them on structured word families, or use prefix-truncation and orbit-advance as floors. His "prefix" quantities are prefix sums K_i and worst-prefix excursions, not truncations of the word to a shorter one. The closed-episode condition of Proposition 2 does not appear.

Parity vectors. For the shortcut map, the bijection between parity vectors of length k and residue classes modulo 2^k is due to Terras [13] (1976) and independently Everett [6] (1977); Lagarias's survey [10] credits both and points to [9, Theorem B] for the precise statement, in the form of the finitely additive invariant measure assigning mass 2^{-n} to each arithmetic progression modulo 2^n. *Verified through the survey, whose full text was consulted; the 1976, 1977 and 1985 originals were not.*

Rozier and Terracol [12] treat parity vectors for the map $n \mapsto (3n+1)/2$ on odd n, with $V_j(n) \in \{0,1\}^j$, introducing a partial order on parity vectors and studying paradoxical sequences. *Full text consulted (v5, 17 May 2026).* That is a different object from the present setting: their parity vectors are binary words of length j with modulus 2^j, whereas a valuation word here records the full exponents $v_2(3x+1)$ and carries modulus 2^{S+1}. They do not treat least elements of such classes.

The leading-run spine. That a leading run of u ones forces $2^{u+1} \mid m+1$, giving $m(1^u) = 2^{u+1} - 1$, is the "modular resistance spine" of Kayadibi [8], who shows $n = 2^m q - 1$ implies $T^j(n) = 3^j 2^{m-j} q - 1$ for $0 \le j \le m-1$ and hence a deterministic lower bound on first-descent time (Definition 3.1, Lemma 3.3, Propositions 3.4 and 3.8). *Full text consulted.* That paper does not treat least realizers or the modulus 2^{S+1}.

Extremal C_w bounds. Sharp bounds on C_w—relevant to the anchor sizes here, though not to anything we prove—are Böhm–Sontacchi Proposition 6, rediscovered as Lemma 4 of Halbeisen and Hungerbühler [7]. *Verified first-hand.*

What the check did not find. No prior treatment was found of: the closed-episode/finite-word distinction of Proposition 2 and its consequence Theorem 5 (c); the exact two-defect closed form (Theorem 3); or the β-reduction of §6. In particular Lagarias's survey [10], which covers the parity-vector framework and its literature, contains no treatment of *least* elements of parity-vector classes, and neither does [1], the one work found to state Proposition 1 independently. This is a negative search result, not a novelty claim, and the audit remains non-exhaustive; but every work cited here has now been read in full or, for three classical items, through Lagarias's survey.

10 Limitations

1. Nothing here bears on the Collatz conjecture; nothing excludes divergent orbits.

2. No formal verification of any kind.
3. Per-configuration $\neq$ uniform; the uniform statement needs Hypothesis β.
4. The residual above F_c is an incomplete-window statement: it requires that $-C_w 3^{-n} \bmod 2^{S+1}$ not be anomalously small. For two defects this is a *two-dimensional* pointwise window problem for $\langle 3 \rangle \leq (\mathbb{Z}/2^M)^*$—simultaneous dyadic windows for 3^{-a} and 3^{-a-b}—whose one-dimensional case is §3. Character and complete-sum methods are provably unavailable there, 2 being a primitive root modulo 3^b, and the two-dimensional case is strictly stronger.
5. Effective methods degrade categorically with a second defect: the form has three logarithms with heights $(3, A, Z)$, $Z = 3^{v+1}(2^{q_a-1} - 1) + 2^{q_a+v}(2^{q_b-1} - 1)$, where $\log_2 Z = \Theta(v)$ exactly and Z is never $\{2, 3\}$-smooth ($19\,773/19\,773$). The recoverable exponent fraction therefore acquires a factor $1/v$ and tends to 0 for every fixed constant, except for bounded middle runs. The three-logarithm constant is **not verified at source**, so no crossover is asserted.
6. The residual behaves as an independent residual. The published statistic—$(\log_2 m - F_c)/|w|$ against P-anchor fullness $\log_2 m(P)/(u + q_a + v + 1)$ at fixed $|w| = 122$—reproduces exactly here, $r = -0.179$, $n = 100$. That design is underpowered: over 40 replicates r ranges over $[-0.372, +0.042]$, sd 0.111, with $|r| > 0.197$ firing in 21/40, so neither magnitude nor sign is stable. At $n = 20\,000$ the same statistic gives $r = -0.1673$, nonzero against the 0.0139 threshold and close to the published estimate. The correlation is real but explains $r^2 = 2.8\%$ of residual variance, so the conclusion holds on grounds of *small effect size*, not statistical insignificance.
7. A literature check was performed (§9) but is not exhaustive; absence from it is not evidence of novelty. Proposition 1 is the author's own prior result ([5]) and is not claimed as new here.

11 Reproduction

All code and data are public at `https://github.com/innerlightr-wq/realizer-floors`. Python 3.14.7, standard library only; no `numpy`, `sympy` or `mpmath`.

```
python3 src/verify_note_numbers.py > data/note_numbers.txt
python3 src_indep/indep_realizer.py          # independent class, R1, denom
python3 src_indep/test_R2_strata.py          # denominator by stratum
python3 src_indep/test_R2_balanced_lift.py   # balanced lift + geom(2/3)
python3 src_indep/test_R3.py                 # F_c validity and dominance
python3 src_indep/test_R4_R5.py              # beta thresholds, merge witnesses
python3 src_indep/test_R5_canonical.py       # canonical merge image
python3 src_indep/test_R6_their_stat.py      # R6 power, published stat
python3 src/two_run_crosscheck.py            # two-run import cross-check

# Appendix A:
python3 src_indep/VERIFY_CONSTRAINED_EXTREMIZER.py   # Thms A.1-A.3, Prop A.5
python3 src_indep/VERIFY_NON_DESCENT.py              # Cor A.7 + certificate
```

Every number in the main text is regenerated by `verify_note_numbers.py`; every number in Appendix A by the two appendix scripts (15/15 and 18/18 checks pass, exact arithmetic). Superseded scripts are retained but not listed: `test_R6_power.py` (a correlation on a different statistic, withdrawn) and `bridge_search_CIRCULAR_retracted.py`.

Open questions

The uniform two-defect floor reduces to a single arithmetic input: **Hypothesis β with $\beta > 1/2$ on the interior family** $U, V \geq 1$ (§6). It is open, and it is an effective question. Everything else proved here is a per-configuration certificate.

Two further items are recorded as limitations rather than results. First, the prior-art check of §9 is not exhaustive: it is a targeted audit of the works nearest to this setting, not a systematic survey, and absence from it is not evidence of novelty. Second, no citation here rests on an abstract alone—every work cited has been read in full, except three classical items cited through Lagarias's survey rather than their originals, as marked.

Provenance

The results were produced twice, by two independently written implementations of the same specification: one via the affine identity of §2–§4, one via a backward recursion on the word that derives the modulus 2^{S+1} with no input from the author. They agree on every shared quantity, including the closed form (0 mismatches out of 19 773) and all floor values. A third, separately produced package was obtained and executed, reproducing its reported correlation $r = -0.179$, merge supremum $+0.3209$, balanced medians $0.337/0.671/1.015$ and $\beta^* = 1/2$.

Three claims made during development were withdrawn and are recorded here because the corrected forms are used above. (i) A bridge-style search reported positive margins by estimating its transfer constant from the measured $m(w)$; that is circular, since the transfer condition rearranges to the conclusion it was meant to establish (§7). (ii) The two representatives of §3 were claimed to differ in height by at most one bit; they differ by one *modulus*, but by up to 9.43 bits of height on the grids tested. (iii) The gain from Theorem 5 (c) was claimed to be $O(1)$; Proposition 6 shows it is unbounded.

References

[1] M. A. K. Akhtar, *Compression Spectra of Collatz Orbits: Sharp Residual Ordering, Exact Valuation Cylinders, and Residue-Threshold Sparsity*, preprint, 19 August 2026, doi:10.5281/zenodo.22015248; SSRN 7314419. *Full text consulted.*

[2] E. De Jesús, *From Sturmian Capacity to Entropy-Deficit Survival: Exact Enumeration and Large Deviations in Confined Accelerated Collatz Words*, v2, 5 October 2026, doi:10.5281/zenodo.23167068. *Full text consulted.*

[3] C. Böhm and G. Sontacchi, Atti Accad. Naz. Lincei Rend. Cl. Sci. Fis. Mat. Natur. **64** (1978), 260–264. *Verified first-hand.*

[4] E. De Jesús, *A Global Occupation Conjecture for the Accelerated* $3x+1$ *Map: Residue–Time Duality, Divergent-Orbit Sparsity, and the Pointwise Realizer Problem*, Revision 8, 4 October 2026, doi:10.5281/zenodo.23146031. *Verified at source.*

[5] E. De Jesús, *The Least-Realizer Minimum for Confined Words: An Erratum to the Terras–Everett Residue Class...*, 11 August 2026, doi:10.5281/zenodo.21889704. *Verified at source.*

[6] C. J. Everett, *Iteration of the number theoretic function* $f(2n) = n$, $f(2n+1) = 3n+2$, Advances in Math. **25** (1977), 42–45. *Cited via [10]; original not consulted.*

[7] L. Halbeisen and N. Hungerbühler, *Optimal bounds for the length of rational Collatz cycles*, Acta Arith. **78** (1997), no. 3, 227–239. *Verified first-hand.*

[8] S. Y. Kayadibi, *A Modular Classification of Pre-Descent Resistance in Accelerated Odd Collatz Dynamics*, 18 June 2026, doi:10.5281/zenodo.20745800. *Full text consulted.*

[9] J. C. Lagarias, *The* $3x+1$ *problem and its generalizations*, Amer. Math. Monthly **92** (1985), 3–23. *Theorem B cited via [10]; original not consulted.*

[10] J. C. Lagarias, *The $3x+1$ problem: an overview*, in: The Ultimate Challenge: The $3x+1$ Problem, Amer. Math. Soc., 2010, pp. 3–29; arXiv:2111.02635. *Full text consulted.*

[11] J. C. Lagarias, *The set of rational cycles for the $3x+1$ problem*, Acta Arith. **56** (1990), 33–53. *Not re-verified at source.*

[12] O. Rozier and C. Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948v5 (17 May 2026); Discrete Math. **349** (2026). *Full text consulted.*

[13] R. Terras, *A stopping time problem on the positive integers*, Acta Arith. **30** (1976), 241–252. *Cited via [10]; original not consulted.*

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A Sharp residual ordering under confinement

Scope. This appendix combines the realizer floors of the main text with the confined-word framework of the companion Sturmian-capacity note [2] and the residual ordering of Akhtar [1]. It proves a sharp constrained optimisation theorem, derives one finite non-descent certificate, and states explicitly why none of this bears on the Collatz conjecture or on EOC.

A.1 Notation, and two deliberate renamings

Throughout $\alpha = \log_2 3$. A valuation word is $w = (d_1, \dots, d_N)$, $d_j \geq 1$, with $S_j = d_1 + \cdots + d_j$, $S_0 = 0$, $S_N = K$, and drift $R_j = S_j - \alpha j$. For an integer corridor $c \geq 0$, w is *c-confined* when $R_j \leq c$ for $1 \leq j \leq N$; equivalently $S_j \leq \lfloor \alpha j + c \rfloor$; equivalently, and this is the form used by every script, $2^{S_j} \leq 2^c 3^j$. Write $\mathcal{C}(N, K, c)$ for the c-confined words of length N with $S_N = K$. The *residual kernel* is

$$H(w) \;=\; \sum_{j=0}^{N-1} \frac{2^{S_j}}{3^j} \;=\; \sum_{j=0}^{N-1} 2^{R_j}.$$

Two renamings avoid collisions. (i) The corridor is c; the main text's closed-episode floor F'_c is written F^{cl} here, so that c is free. (ii) The companion note uses $\beta = \alpha - 1$; the main text uses β for the open exponent hypothesis of §6. This appendix writes $\alpha - 1$ in full and never uses β. Akhtar's kernel H is unrelated to the binary entropy H_2 of the companion note; the two never occur together.

A.2 The constrained extremizers

Theorem A.1 (Constrained prefix-sum dominance)**.** Let $N \geq 1$, let $c \geq 0$ be an integer, and let K satisfy $N \leq K \leq \lfloor \alpha N + c \rfloor$. Put

$$S_j^* = \min\bigl\{ \lfloor \alpha j + c \rfloor,\; K - N + j \bigr\} \;(1 \leq j \leq N), \qquad S_0^* = 0, \qquad d_j^* = S_j^* - S_{j-1}^*.$$

Then *(i)* every $d_j^* \geq 1$, so w^* is a valuation word; *(ii)* $\sum_j d_j^* = K$; *(iii)* $w^* \in \mathcal{C}(N, K, c)$; and *(iv)* every $w \in \mathcal{C}(N, K, c)$ satisfies $S_j \leq S_j^*$ for all j.

Proof. (iv) Confinement gives $S_j \le \lfloor \alpha j + c \rfloor$. The remaining $N - j$ letters are each at least 1, so $K - S_j \ge N - j$, i.e. $S_j \le K - N + j$. Take the minimum. (i) Each argument of the minimum increases by at least 1 as j increases by 1 — for the first because $\lfloor \alpha j + c \rfloor - \lfloor \alpha(j-1) + c \rfloor \ge \lfloor \alpha \rfloor = 1$, for the second exactly — and if $f(j) \ge f(j-1) + 1$ and $g(j) \ge g(j-1) + 1$ then $\min\{f, g\}$ inherits the same increment bound. (ii) $S_N^* = \min\{\lfloor \alpha N + c \rfloor, K\} = K$ by hypothesis. (iii) Immediate from the definition. □

Theorem A.2 (Sharp maximum)**.** For every $w \in \mathcal{C}(N, K, c)$,

$$H(w) \;\le\; H(w^*) \;=\; 1 + \sum_{j=1}^{N-1} \frac{2^{\min\{\lfloor \alpha j + c \rfloor,\, K - N + j\}}}{3^j} \;=:\; H_{\max}^{\mathrm{conf}}(N, K, c),$$

with equality if and only if $w = w^*$.

Proof. H depends only on $S_0, \ldots, S_{N-1}$ and is strictly increasing in each separately. Theorem A.1(iv) bounds each S_j by S_j^*, and (i)–(iii) show the bound is attained simultaneously by the admissible word w^*. Equality forces $S_j = S_j^*$ for $j \le N - 1$ by strict monotonicity; with $S_N = K$ this determines every d_j, so $w = w^*$. Note $d_N = K - S_{N-1}^* \ge 1$ because $S_{N-1}^* \le K - 1$. □

Theorem A.3 (Sharp minimum)**.** If $\mathcal{C}(N, K, c) \ne \emptyset$ then $w_{\min} = (1, \ldots, 1, K - N + 1)$ lies in it, and for every $w \in \mathcal{C}(N, K, c)$

$$H(w) \;\ge\; H(w_{\min}) \;=\; \sum_{j=0}^{N-1} \Big(\tfrac{2}{3}\Big)^j \;=\; 3\Big(1 - \big(\tfrac{2}{3}\big)^N\Big),$$

with equality iff $w = w_{\min}$. The value depends on neither K nor c.

Proof. The prefix sums of $w_{\min}$ are $S_j = j$ for $j < N$ and $S_N = K$; since $\alpha > 1$ and $c \ge 0$ we have $\lfloor \alpha j + c \rfloor \ge j$, and $K \le \lfloor \alpha N + c \rfloor$ by hypothesis, so $w_{\min}$ is c-confined. Any valuation word has $S_j \ge j$, hence $2^{S_j} \ge 2^j$ and $H(w) \ge \sum_{j<N} (2/3)^j$. Equality forces $S_j = j$ for $j \le N - 1$, i.e. $d_1 = \cdots = d_{N-1} = 1$. □

Corollary A.4 (Sharp residual interval)**.** For every $w \in \mathcal{C}(N, K, c)$, $3\big(1 - (2/3)^N\big) \le H(w) \le H_{\max}^{\mathrm{conf}}(N, K, c)$, both endpoints attained, each by a unique word.

A.3 Relation to the unrestricted ordering

Akhtar's unrestricted maximiser over all words with fixed (N, K) is the front-loaded word $(K - N + 1, 1, \ldots, 1)$ [1, Thm. 6]. It concentrates the entire excess valuation in the first letter, so its drift $R_1 = K - N + 1 - \alpha$ is as large as the family allows — which is exactly what every-prefix confinement forbids.

Proposition A.5 (When confinement bites)**.** The constrained maximiser coincides with the unrestricted front-loaded maximiser if and only if that word is itself c-confined, i.e. iff $K - N + 1 \le \lfloor \alpha + c \rfloor$. At $c = 0$, $\lfloor \alpha \rfloor = 1$, so this holds iff $K = N$ — the degenerate all-ones family. Otherwise the constrained maximum is strictly smaller.

Proof. If the front-loaded word is confined it lies in $\mathcal{C}(N, K, c)$ and maximises H over the larger unrestricted family, hence over $\mathcal{C}$; Theorem A.2 identifies it with w^*. Conversely w^* is confined by Theorem A.1(iii). The displayed inequality is the $j = 1$ confinement condition; the later ones follow because the front-loaded prefix sums increase by exactly 1 after $j = 1$ while $\lfloor \alpha j + c \rfloor$ increases by at least 1. □

Proposition A.6 (The elementary bound is already order-sharp)**.** From $R_j \le c$ one gets $H(w) \le N2^c$ immediately. The exact maximiser does *not* improve the order. Measured values of $H_{\max}^{\mathrm{conf}}/(N2^c)$: 0.763, 0.748, 0.735 at $(N, c) = (8, 0)$, $(16, 0)$, $(32, 0)$; and 0.700–0.720 at $c = 1$ for $N = 8, \ldots, 64$. Both quantities are $\Theta(N)$ at fixed c. The sharp theorem buys a bounded constant factor here, not a better exponent.

A.4 Positive-compression thresholds, and their scope

With $C(w) = \sum_{j=0}^{N-1} 3^{N-1-j} 2^{S_j}$ the accelerated identity is $m_N = (3^N m_0 + C(w))/2^K$, and $C(w) = 3^{N-1} H(w)$.

Corollary A.7 (Threshold form)**.** Suppose $2^K > 3^N$, equivalently $R_N > 0$. Then

$$T(w) \;=\; \frac{C(w)}{2^K - 3^N} \;=\; \frac{H(w)}{3\big(2^{R_N} - 1\big)}, \qquad \text{hence} \qquad T(w) \;\le\; \frac{H_{\max}^{\mathrm{conf}}(N, K, c)}{3\big(2^{R_N} - 1\big)}$$

for every $w \in \mathcal{C}(N, K, c)$ with $R_N > 0$.

Proof. $2^K - 3^N = 3^N(2^{K-\alpha N} - 1) = 3^N(2^{R_N} - 1)$; divide $C = 3^{N-1}H$ by it and apply Theorem A.2. □

Scope restrictions, preserved throughout. The hypothesis $2^K > 3^N$ is essential: T is meaningless otherwise. *At $c = 0$ no confined word has positive terminal compression*, since $K \le \lfloor \alpha N \rfloor$ forces $2^K \le 3^N$ (verified on 31 cases, 0 violations); the $c = 0$ example below therefore tests the *ordering* theorem, not non-descent. For $c > 0$ positive compression requires $0 < R_N \le c$. Finally, Corollary A.7 bounds *one word's* threshold; it is not a statement about least realizers, and not an assertion that a non-descent witness exists.

A.5 A finite exclusion certificate

Akhtar's criterion requires a positive-compression non-descent witness to satisfy $m \le T(w)$ [1, Prop. 13]. The main text gives $m(w) \ge F^{\mathrm{cl}}(w)$ for a two-defect word. Combining,

$$\boxed{F^{\mathrm{cl}}(w) > T(w) \implies w \text{ has no positive-compression non-descent witness.}}$$

This is an immediate combination of two established inequalities, not a new theorem; we label it as such. A genuinely new result would be a proved condition on (u, q_a, v, q_b, t) guaranteeing it without computing $m(w)$, and none is claimed.

A.6 Two independently checked examples

(a) Sharp ordering, $N = 8$, $K = 12$, $c = 0$. $w^* = (1, 2, 1, 2, 1, 2, 2, 1)$ with prefix sums $(1, 3, 4, 6, 7, 9, 11, 12)$ and

$$H_{\max}^{\mathrm{conf}} = \tfrac{13349}{2187} = 6.1037951532\ldots$$

exactly, as a rational. Of the 330 compositions of 12 into 8 positive parts, exactly 85 are every-prefix 0-confined. The unrestricted front-loaded word $(5, 1, 1, 1, 1, 1, 1, 1)$ has $H = 31.127\ldots$ and is *not* 0-confined, illustrating Proposition A.5. Since $c = 0$, this is a test of the ordering theorem only.

(b) Exclusion certificate, $w = (1, 2, 1, 3, 1)$**.** Here $u = 1, q_a = 2, v = 1, q_b = 3, t = 1$, so $N = 5$, $K = 8$, prefix sums $(1, 3, 4, 7, 8)$. The word is 1-confined but not 0-confined, and $2^K - 3^N = 256 - 243 = 13 > 0$, so the positive-compression regime applies. Exactly:

$$C(w) = 383 = 3^4 H(w), \qquad T(w) = \tfrac{383}{13} = 29.4615\ldots$$

and the identity of Corollary A.7 is confirmed. The floors are $F_c(w) = 27$ and $F^{\mathrm{cl}}(w) = 59$, with true least realizer $m(w) = 187$. Thus $F^{\mathrm{cl}}(w) = 59 > T(w)$ *excludes* a positive-compression non-descent witness, whereas $F_c(w) = 27 < T(w)$ does *not*: the closed-episode improvement of the main text is what delivers this certificate. Consistently, $m(w) = 187 > T(w)$.

A.7 Limitations, provenance, and why this does not touch EOC

(1) **A restarted divergent tail is 0-confined.** A hypothetical divergent orbit restarted after its final drift maximum has $R_j \le 0$ at every depth. By the scope restriction above, the positive-compression threshold is then *unavailable* — the entire mechanism of Corollaries A.7–A.7 is vacuous exactly on the class that would matter for EOC.

(2) **Finite realization is not infinite realization.** Every prefix of a word is realized by some positive integer; that one fixed integer realizes the whole infinite word is a different, stronger statement.

(3) **Population sparsity is not pointwise placement.** A density-zero exceptional set may contain every least-realizer champion.

(4) **Sharper finite-window estimates do not control separated returns.**

Attribution. The accelerated identity and the parity-vector correspondence are classical (see the main text's prior-work section). The residual ordering, the unrestricted extremizers, the exact valuation-word residue class and the non-descent criterion are Akhtar's [1]. The confined-word/Sturmian framework and the entropy deficit are the author's companion note [2]. The closed-episode floor F^{cl} is the main text. New here are Theorems A.1–A.3, Corollary A.4 and Proposition A.5; everything in §E.5 is a labelled combination.